Area, Surface Area & Volume Error Analysis TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Area of a triangle

ORIGINAL PROBLEM

Find the area of a triangle with base 10 and height 6.

INCORRECT WORKED SOLUTION

$A = bh$

$A = 10 \cdot 6 = 60$

A. EXPLAIN THE MISTAKE (1 POINT)

A triangle is HALF of a rectangle: $A = \frac{1}{2}bh = \frac{1}{2}(10)(6) = 30$.

B. SHOW CORRECT WORK (1 POINT)

$A = \frac{1}{2}bh$

$A = \frac{1}{2}(10)(6) = 30$

2. Area of a parallelogram

ORIGINAL PROBLEM

A parallelogram has base 8, slanted side 5 and height 4. Find its area.

INCORRECT WORKED SOLUTION

$A = \text{base} \cdot \text{side}$

$A = 8 \cdot 5 = 40$

A. EXPLAIN THE MISTAKE (1 POINT)

Area uses the perpendicular HEIGHT, not the slanted side: $A = 8 \cdot 4 = 32$.

B. SHOW CORRECT WORK (1 POINT)

$A = \text{base} \cdot \text{height}$

$A = 8 \cdot 4 = 32$

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3. Area of a trapezoid

ORIGINAL PROBLEM

Find the area of a trapezoid with bases 6 and 10 and height 4.

INCORRECT WORKED SOLUTION

$A = (b_1 + b_2)h$

$A = (6 + 10)(4) = 64$

A. EXPLAIN THE MISTAKE (1 POINT)

The trapezoid formula includes $\frac{1}{2}$: $A = \frac{1}{2}(b_1 + b_2)h = \frac{1}{2}(16)(4) = 32$.

B. SHOW CORRECT WORK (1 POINT)

$A = \frac{1}{2}(b_1 + b_2)h$

$A = \frac{1}{2}(6 + 10)(4)$

$A = 32$

4. Shaded regions

ORIGINAL PROBLEM

A circle with diameter 10 is cut out of a 10 by 10 square. Find the area of the shaded region.

INCORRECT WORKED SOLUTION

Square: $10 \cdot 10 = 100$

Circle: $\pi(10)^2 = 100\pi$

Shaded $= 100 - 100\pi$

A. EXPLAIN THE MISTAKE (1 POINT)

The circle's radius is 5 (half the diameter), so its area is $\pi(5)^2 = 25\pi$. Shaded $= 100 - 25\pi \approx 21.5$.

B. SHOW CORRECT WORK (1 POINT)

Square: 100

Circle: $r = 5$, $A = \pi(5)^2 = 25\pi$

Shaded $= 100 - 25\pi \approx 21.5$

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5. Volume of a cylinder

ORIGINAL PROBLEM

Find the volume of a cylinder with diameter 6 and height 7.

INCORRECT WORKED SOLUTION

$V = \pi r^2 h$

$V = \pi(6)^2(7)$

$V = 252\pi$

A. EXPLAIN THE MISTAKE (1 POINT)

The radius is half the diameter: r = 3. $V = \pi(3)^2(7) = 63\pi$.

B. SHOW CORRECT WORK (1 POINT)

$r = 3$

$V = \pi(3)^2(7)$

$V = 63\pi$

6. Volume of a cone

ORIGINAL PROBLEM

Find the volume of a cone with radius 3 and height 4.

INCORRECT WORKED SOLUTION

$V = \pi r^2 h$

$V = \pi(3)^2(4)$

$V = 36\pi$

A. EXPLAIN THE MISTAKE (1 POINT)

A cone holds ONE-THIRD as much as a cylinder: $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(9)(4) = 12\pi$.

B. SHOW CORRECT WORK (1 POINT)

$V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3}\pi(3)^2(4)$

$V = 12\pi$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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7. Height vs. slant height

ORIGINAL PROBLEM

A cone has radius 6 and slant height 10. Find its volume.

INCORRECT WORKED SOLUTION

$V = \frac{1}{3}\pi r^2 h$

$V = \frac{1}{3}\pi(6)^2(10)$

$V = 120\pi$

A. EXPLAIN THE MISTAKE (1 POINT)

The formula needs the perpendicular HEIGHT, not the slant height. $h = \sqrt{10^2 - 6^2} = 8$, so $V = \frac{1}{3}\pi(36)(8) = 96\pi$.

B. SHOW CORRECT WORK (1 POINT)

$h^2 + 6^2 = 10^2$, so $h = 8$

$V = \frac{1}{3}\pi(6)^2(8)$

$V = 96\pi$

8. Volume of a sphere

ORIGINAL PROBLEM

Find the volume of a sphere with radius 3.

INCORRECT WORKED SOLUTION

$V = \frac{4}{3}\pi r^3$

$V = \frac{4}{3}\pi(3)^2$

$V = 12\pi$

A. EXPLAIN THE MISTAKE (1 POINT)

The radius is CUBED: $3^3 = 27$. $V = \frac{4}{3}\pi(27) = 36\pi$.

B. SHOW CORRECT WORK (1 POINT)

$V = \frac{4}{3}\pi r^3$

$V = \frac{4}{3}\pi(3)^3$

$V = \frac{4}{3}\pi(27) = 36\pi$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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9. Surface area of a prism

ORIGINAL PROBLEM

Find the surface area of a rectangular prism that is 3 by 4 by 5.

INCORRECT WORKED SOLUTION

$2(3 \cdot 4) = 24$

$2(4 \cdot 5) = 40$

$SA = 24 + 40 = 64$

A. EXPLAIN THE MISTAKE (1 POINT)

A rectangular prism has THREE pairs of faces. The top and bottom, $2(3 \cdot 5) = 30$, were left out. $SA = 24 + 40 + 30 = 94$.

B. SHOW CORRECT WORK (1 POINT)

$2(3 \cdot 4) = 24$

$2(4 \cdot 5) = 40$

$2(3 \cdot 5) = 30$

$SA = 24 + 40 + 30 = 94$

10. Square units

ORIGINAL PROBLEM

A rectangle is 2 ft by 3 ft. Find its area in square inches.

INCORRECT WORKED SOLUTION

$\text{Area} = 2 \cdot 3 = 6 \text{ ft}^2$

$1 \text{ ft} = 12 \text{ in}$

$6 \cdot 12 = 72 \text{ in}^2$

A. EXPLAIN THE MISTAKE (1 POINT)

Square units need the conversion factor SQUARED: $1 \text{ ft}^2 = 12 \cdot 12 = 144 \text{ in}^2$. So $6 \cdot 144 = 864 \text{ in}^2$.

B. SHOW CORRECT WORK (1 POINT)

$\text{Area} = 6 \text{ ft}^2$

$1 \text{ ft}^2 = 12^2 = 144 \text{ in}^2$

$6 \cdot 144 = 864 \text{ in}^2$

Check: $24 \text{ in} \cdot 36 \text{ in} = 864 \text{ in}^2$